

Newton over Hodge at $p = 2$ for 2-power-order characters on arbitrary smooth affine curves

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ABSTRACT. Newton-over-Hodge lower bounds for Artin–Schreier–Witt L -functions on arbitrary smooth affine curves are known only for odd p : at $p = 2$ the Kramer–Miller–Upton local estimate degrades to $a(k) = \lfloor (k-1)/3 \rfloor$, which they call too low for applications to the global setting. We show that this is an artifact of one weight choice, not of the prime. The 3 is the tame ramification index of an auxiliary Belyi map, and the weight $a(k) = \lfloor (k-1)/3 \rfloor + (k \bmod 2)$ is admissible for every k , with defect $d(k) \geq \max(1, k/6)$; the rate $1/6$ is exactly optimal, since $k = 2e$ is a self-loop of the local operator with coefficient 2 for every weight. Adding a characteristic-2 tame Belyi map of uniform index 3 over the point 1, we obtain $\mathrm{NP}_q(L(\rho, s)) \geq \mathrm{HP}_q(\rho)$ as full polygons, with no truncation, for every nontrivial character of 2-power order on a smooth affine curve over $\mathbf{F}_{2^a}$. We also give the local-to-global contact criterion at $p = 2$ on an initial segment, prove that restriction structural, and record corrections to the literature.

1. INTRODUCTION

1.1. The landscape. Let p be a prime, q a power of p , and $X/\mathbb{F}_q$ a smooth affine curve with smooth compactification $\overline{X}$ of genus g and $S = \overline{X} \setminus X$. For a nontrivial finite character ρ of $\pi_1(X)$, the L -function $L(\rho, s)$ is a polynomial, and the Newton-over-Hodge problem asks for a lower bound on its q -adic Newton polygon $\mathrm{NP}_q(\rho)$ in terms of the ramification of ρ alone.

Over the projective line and its affinoids the problem is closed at every prime, $p = 2$ included. Zhu’s theorem on one-dimensional affinoids [26] is sharp for any prime coprime to the pole orders; Liu and Wan’s T -adic bound [18, Theorem 5.2] carries no hypothesis on p whatsoever, and neither does Schmidt’s affinoid version [22]; the Davis–Wan–Xiao tower [7] needs only $\gcd(d, p) = 1$. Nothing below is a claim about $\mathbb{P}^1$. Behind these sit the Newton-polygon theory of exponential sums of Adolphson and Sperber [1, 2] and Wan [24, 25], and the stratification results of Blache and Férard [3]; the T -adic line continues in [10].

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Over an *arbitrary* smooth affine curve the situation is different, and the difference is the point of this paper. The framework that produces a *ramification-defined* Hodge bound there — one built from the Swan conductors of ρ and nothing else — is Kramer-Miller’s and Kramer-Miller-Upton’s. It comprises [15] for exponential sums, [14] for abelian Artin L -functions, and [16, 17] for the local-to-global contact theory and its variation in families. Every one of these papers assumes p odd. The hypothesis is printed in the first section of each: “*Let p be a prime with $p \geq 3$* ” [14], “*Let p be an odd prime and let q be a power of p* ” [16].

The exclusion is neither an oversight nor an unattempted case. The authors name its two components.

1.2. The obstruction, in the authors’ words. The first component is analytic. At a point P where the auxiliary Belyi map η takes the value 1, the relevant Frobenius is not a p -power map, and the Dwork operator U_p moves poles around in a way that has to be controlled by a weight on a growth module. Kramer-Miller and Upton carried out the $p = 2$ computation and recorded the outcome in their Remark 6.5, which we quote in full because it is the statement this paper repairs.

Suppose that $p = 2$. For $k \geq 3$, define $a(k) = \lfloor (k-1)/3 \rfloor$. A similar construction provides a submodule $\mathcal{A}_{\pi,P}^{\mathbf{m},*} \subset A_{\pi,P}^{\dagger}$ with the following property: Let $k = 2\ell - r$ with $r = 0$ or 1 . Then

$$U_p(\pi^{a(k)/m_P} t_P^{-k}) \in \pi^{(a(k)-a(\ell+r))/m_P} \mathcal{A}_{\pi,P}^{\mathbf{m},*}.$$

This estimate is too low for applications to the global setting. For example, if $k = 5 = 2 \cdot 3 - 1$, then $a(k) - a(\ell + r) = 0$, and this contributes an extra segment of slope 0 in the global Hodge bound below. [16, Remark 6.5]

The estimate is not merely weak: it breaks a *count*. The Deuring–Shafarevich argument that supplies the slope-0 part of the polygon pins the number of slope-0 segments exactly, and a single index with zero gain inserts one more. That is why the remark says “too low for applications” rather than “weaker than we would like”.

The second component is geometric, and was stated two years earlier:

Finally, we mention our requirement that $p \geq 3$. When $p = 2$ it is likely that the methods in this paper still work. The main difficulty is that some estimates in section 4 must be modified. It is also not immediately clear that we can find a cover $\eta : X \rightarrow \mathbb{P}_{\mathbb{F}_q}^1$ satisfying the desired properties. To construct η , we use the fact that X admits a simply branched map to $\mathbb{P}_{\mathbb{F}_q}^1$, which is false when $p = 2$. [15, §1.4]

The construction that fails is a composition ending in the $(p-1)$ -power map $z \mapsto z^{p-1}$, whose job is to create a fibre of uniform ramification index $p-1$ over the point 1. At $p = 2$ that map is the identity, and the whole stage evaporates.

The gap is live rather than historical. Booher, Groen and Kramer-Miller [5] treat $\mathbb{Z}/2$ -covers in characteristic two through their Ekedahl–Oort types rather than the full 2-adic Newton polygon, which is the move one makes when the 2-adic polygon is out of reach.

1.3. The result. The obstruction of Remark 6.5 is not analytic. The number 3 in $\lfloor (k-1)/3 \rfloor$ is the tame ramification index of the auxiliary map, and the estimate is repaired by a different weight on the same module — Kramer-Miller–Upton’s own weight plus a parity indicator,

$$a(k) = \left\lfloor \frac{k-1}{3} \right\rfloor + (k \bmod 2),$$

which we prove admissible for every k , with defect $d(k) \geq 1$ and $d(k) \sim k/6$, that rate being exactly optimal (§2–§3). Together with a characteristic-2 tame Belyi map of uniform index 3 over the point 1 (§4), this yields the following.

Theorem 1.1 (Main theorem). *Let $q = 2^a$, let $X/\mathbb{F}_q$ be a smooth affine curve with smooth compactification $\overline{X}$ of genus g and $S = \overline{X} \setminus X$, and let ρ be a nontrivial finite character of $\pi_1(X)$ of any 2-power order 2^n . Then*

$$\mathrm{NP}_q(\rho) \succeq \mathrm{HP}_q(\rho)$$

as full polygons: with the ramification-defined Hodge polygon of [16, §1.2], no truncation, and no restriction on n .

Here $\mathrm{HP}_q(\rho)$ is the Kramer-Miller ramification-defined Hodge polygon, the polygon with slope multiset

$$(1) \quad \{0\}^{g-1+|S|} \cup \{1\}^{g-1+|S|} \cup \bigcup_{P \in S} \left\{ \frac{1}{d_P}, \frac{2}{d_P}, \dots, \frac{d_P-1}{d_P} \right\},$$

where d_P is the Swan conductor of ρ at P and $|S|$ is counted geometrically; $\succeq$ is the relation “lies on or above”, as in the sources. Theorem 1.1 is the exact characteristic-2 analogue of [14, Theorem 1.1] for characters of p -power order, with the same polygon and the same generality of base.

The local-to-global *contact* criterion of [16, Theorem 1.1] — the two polygons touch globally if and only if they touch locally at every wild point — also goes through, but only on an initial segment.

Theorem 1.2 (Contact, on an initial segment). *Assume X ordinary and let ρ have order 2^n . For every q -adic r with $0 \leq r \leq 2^{1-n}$, the polygons $\mathrm{HP}_q^{<r}(\rho)$ and $\mathrm{NP}_q^{<r}(\rho)$ have the same terminal point if and only if $\mathrm{HP}_q^{<r}(\rho_P^{\mathrm{ext}})$ and $\mathrm{NP}_q^{<r}(\rho_P^{\mathrm{ext}})$ have the same terminal point for every $P \in S$. In particular the full range $r \in [0, 1]$ of [16, Theorem 1.1] is covered at order 2.*

The restriction in Theorem 1.2 is not an artifact of our route. We prove in §3 that the index $k = 2e$ is a self-loop of the type-2 operator with transition coefficient exactly 2, for *every* odd tame index e and independently of e ; so $d(2e) \leq 1$ for every weight whatsoever, and no choice of auxiliary index

relieves the cap. Theorem 1.2 rests in addition on an exactness statement that [16] asserts without proof, at every p and for its own weight; Theorem 1.1 does not (§5).

1.4. Architecture of the proof. The global machinery is entirely Kramer–Miller’s. What this paper supplies is the pair of ingredients their own papers name as missing, and the verification that nothing else in the argument sees the prime.

- (1) **The type-2 operator, solved (§2).** Writing $U_2(t^{-k}) = \sum_j c_{k,j} t^{-j}$ at odd tame index e , the support is a single arithmetic progression $j'(k) + e\mathbb{Z}_{\geq 0}$ and the coefficients along it are the Taylor coefficients of $\cosh(\lambda \operatorname{arcsinh} z)$ and $\sinh(\lambda \operatorname{arcsinh} z)/z$ with $\lambda = k/e$ (Theorem 2.2); the proof is a change of variable turning the defining trace identity into one second-order ODE.
- (2) **Their valuations, exactly.** Legendre’s formula converts the closed form into an exact expression for $v_2(c_{k,j})$ (Theorem 2.5), whence a four-case parity argument gives the tail estimate $v_2 \geq m$, tight only at $k \equiv 2 \pmod 4$, $m = 1$ (Lemma 2.6).
- (3) **The repaired weight.** That tightness and the failure of $\lfloor (k-1)/3 \rfloor$ occur at the same indices and in opposite directions, which is why a parity indicator — and nothing more — fixes the estimate (Theorem 2.8).
- (4) **Sharpness (§3).** One coefficient, $c_{2e,2e} = 2$, caps every weight at rate $1/(2e)$; at $e = 3$ that is $1/6$, which the repaired weight attains.
- (5) **The geometric input (§4).** Replacing the $(p-1)$ -power stage by an e -power stage, and Fulton’s simply-branched map by the tame map of Kedlaya–Litt–Witaszek and Sugiyama–Yasuda, produces the required Belyi map at $p = 2$ (Lemma 4.1).
- (6) **Assembly (§5).** The two ingredients go into [14], whose remaining hypotheses are p -uniform; the weight transports through a dictionary between the two papers’ growth modules that holds on the nose.

En route we record four corrections to the sources, none of which is a $p = 2$ obstruction and none of which invalidates a published theorem: a paraphrase in [16] that over-states the freedom in choosing a linear map (§4.3), a strict inequality in [16, Definition 6.3] inconsistent with its own claim, and two statements in [14, §4.1] that are false as printed and true in a repaired form (§5.4). All four are p -uniform.

1.5. Provenance, machine verification, and what is not claimed.

This paper was produced by a multi-agent artificial-intelligence workflow under human direction. Separate agents extracted and quoted the sources, attacked the problem, adversarially audited the results, and drafted the exposition; the audit was run as a hostile re-derivation rather than a review, and it refuted or downgraded several intermediate claims, including one range statement that had already propagated into a dependent draft. Every theorem stated here was re-derived independently of the draft that first

proposed it, and every computational assertion is backed by at least two implementations sharing no code, in exact rational arithmetic.

The computational substrate is AXEYUM [4], an open-source Rust automated-reasoning stack whose stated discipline — untrusted fast search, trusted small checking — is the one this section describes. Every computation reported below was performed in its proof-carrying computer-algebra layer as a self-checking program, at a public pinned commit.¹ The second implementation of each computation is deliberately *not* built on that stack: it is a from-scratch reimplementation in a different language, sharing no line of code, whose purpose is to disagree if the first is wrong. The replication package runs both layers in that order — each asserting its findings and exiting nonzero on failure — together with the complete research log and an error ledger recording what the project got wrong before it got it right.

The proofs in this paper have not yet been read by a human referee. We state that once, here, rather than qualifying each theorem: the distinction that matters to a reader is not which statement carries which internal label, but that the whole has had machine verification and adversarial re-derivation and has not had peer review. Every quotation from a source was transcribed from the published or arXiv text by at least two independent readers; the passage in §1.2 was transcribed five times.

We claim nothing for Newton over Hodge at $p = 2$ over $\mathbb{P}^1$ or over affinoids, which is published (§1.1). We do not re-verify Deuring–Shafarevich, Katz–Gabber, Monsky’s trace formula, the functional analysis of [14, §6], or the internal proofs of [13] and [23]; these are cited as their users cite them. Section 6 states what remains open.

1.6. Organisation. Section 2 solves the type-2 operator and proves the repaired weight admissible. Section 3 shows the rate is exactly $1/6$ and draws the consequence for the contact theory. Section 4 constructs the characteristic-2 Belyi map and settles the arithmetic condition it needs. Section 5 assembles Theorem 1.1 and records the source defects met on the way. Section 6 states the open problems.

2. THE TYPE-2 OPERATOR AT $p = 2$

Throughout this section $p = 2$ and e is an odd integer. Nothing here uses $e = 3$ except Theorem 2.8, whose case analysis is modulo 6; the closed form, the valuation identity and the tail estimate hold for every odd e .

¹<https://github.com/mjbommar/axeyum>, branch `agent/noh-p2-axeyum-examples`, commit `75663ef85c2dad4390a3b6d77361919a914642a9`. The two programs this paper depends on are `crates/axeyum-cas/examples/noh_u2_matrix.rs` — the type-2 operator computed exactly and anchored to the Dwork trace formula against point counts over $\mathbb{F}_{2^k}$ — and `noh_wt_certificate.rs`, the certificate for Theorems 2.2, 2.5, 2.8 and 3.1 and Lemma 2.6. Both assert their own findings and exit nonzero on failure; the replication package verifies by SHA-256 that the copies it ships are byte-identical to that commit before running them.

2.1. The operator and its support. Let $\eta : X \rightarrow \mathbb{P}^1$ be a tame Belyi map and let P be a point with $\eta(P) = 1$ and ramification index e . Write $t = t_P$ for a local parameter at P and $u = t^e$ for the pullback of the parameter at 1 downstairs. The local Frobenius downstairs is $\sigma(u) = (u + 1)^p - 1$ [16, §4.3], [14, §3.4]. We follow [16] in writing σ for the p -Frobenius; [14] and [15] call the same map ν and reserve σ for its a -th iterate, so a display compared across the two papers must be read with that in mind. At $p = 2$,

$$\sigma(t)^e = \sigma(u) = u^2 + 2u = t^{2e} + 2t^e,$$

so that, writing $x := 1/t$,

$$\sigma(t) = t^2 G, \quad G := (1 + 2x^e)^{1/e}.$$

The binomial series G converges because $\gcd(e, 2) = 1$; this is the only place oddness of e is used, and it is used again nowhere. We have $[\mathcal{E}^\dagger : \sigma(\mathcal{E}^\dagger)] = 2$, and the nontrivial conjugate of t over $\sigma(\mathcal{E}^\dagger)$ is $t' = -tG$: the two roots of $Y^2 + 2Y - \sigma(u)$ are u and $u' = -2 - u$, and for e odd

$$(-tG)^e = -t^e(1 + 2x^e) = -(u + 2) = u'.$$

Let $U_p = \frac{1}{p} \sigma^{-1} \circ \text{Tr}_{\mathcal{E}^\dagger/\sigma(\mathcal{E}^\dagger)}$ be the Dwork operator of [16, Definition 4.9]. Because $\eta(P) = 1$ forces $P \notin S$, the character ρ is unramified at P and *no splitting function enters this estimate at all*; the constant $\tilde{E}_P$ of [16, Proposition 6.4] lies in $1 + \pi R_q$ and is invisible to every valuation below. Writing

$$(2) \quad U_2(t^{-k}) = \sum_j c_{k,j} t^{-j}$$

and applying σ to both sides, using $\text{Tr}(t^{-k}) = x^k(1 + (-1)^k G^{-k})$ and $\sigma(t^{-j}) = x^{2j} G^{-j}$, gives the defining identity

$$(3) \quad \frac{1}{2} x^k (1 + (-1)^k G^{-k}) = \sum_j c_{k,j} x^{2j} G^{-j}.$$

Lemma 2.1 (Support). *The coefficients $c_{k,j}$ vanish unless $j \in j'(k) + e\mathbb{Z}_{\geq 0}$, where*

$$j'(k) = \frac{k}{2} \quad (k \text{ even}), \quad j'(k) = \frac{k+e}{2} \quad (k \text{ odd}).$$

Proof. G is a series in x^e , so the left side of (3) lives in degrees congruent to k modulo e and the j -th term on the right in degrees congruent to $2j$; as e is odd, 2 is invertible modulo e and the admissible j form a single class. For the minimal degree: if k is even then $1 + G^{-k} = 2 - (2k/e)x^e + \dots$, so the left side starts at x^k and $2j = k$; if k is odd then $1 - G^{-k} = (2k/e)x^e + \dots$, so it starts at x^{k+e} and $2j = k + e$, an integer because k and e are both odd. $\square$

At $e = 3$ this is exactly the index bookkeeping of [16, Remark 6.5]: with $k = 2\ell - r$ and $r \in \{0, 1\}$, $k = 2c$ gives $\ell + r = c = j'(k)$ and $k = 2c - 1$ gives $\ell + r = c + 1 = j'(k)$. It is worth noting that the fit is exact only at

$e = 3$: at $e = 5$ the same computation does not fit the shape $k = 2\ell - r$ with $r \in \{0, 1\}$, so the tame index 3 is singled out by Kramer-Miller-Upton's own phrasing rather than merely consistent with it.

2.2. The closed form.

Theorem 2.2 (Closed form). *Let $p = 2$ and e odd. For every $k \geq 1$ and $m \geq 0$,*

$$(4) \quad c_{k, k/2+em} = \frac{\prod_{i=0}^{m-1} (k^2 - 4e^2 i^2)}{e^{2m} (2m)!} \quad (k \text{ even}),$$

$$(5) \quad c_{k, (k+e)/2+em} = \frac{k}{e} \cdot \frac{\prod_{i=0}^{m-1} (k^2 - e^2 (2i+1)^2)}{e^{2m} (2m+1)!} \quad (k \text{ odd}).$$

Proof. Put $v := 2x^e$, so that $G = (1+v)^{1/e}$ and $G^e = 1+v$; put $(1+v)^{1/2} = \exp(\varphi)$, that is $\varphi = \frac{1}{2} \log(1+v)$, and

$$W := \frac{x^{2e}}{1+2x^e} = \frac{(v/2)^2}{1+v} = \frac{v^2}{4(1+v)}.$$

Then $W = \sinh^2(\varphi)$, since $\sinh(\varphi) = \frac{1}{2}((1+v)^{1/2} - (1+v)^{-1/2}) = v/(2(1+v)^{1/2})$. Note that W does not depend on e . Put $\tau := \varphi/e$, so that $(1+v)^{k/(2e)} = \exp(k\tau)$ and

$$(6) \quad \sinh(e\tau) = \sinh(\varphi) = x^e(1+v)^{-1/2}.$$

Divide (3) by $x^{2j'} G^{-j'}$ with $j' = j'(k)$. Using $G^e = 1+v$, the right side becomes

$$\sum_{m \geq 0} c_{k, j'+em} x^{2em} G^{-em} = \sum_{m \geq 0} c_{k, j'+em} \left(\frac{x^{2e}}{1+v} \right)^m = \sum_{m \geq 0} c_{k, j'+em} W^m.$$

The left side becomes, for k even with $j' = k/2$,

$$\frac{\frac{1}{2} x^k (1 + G^{-k})}{x^k G^{-k/2}} = \frac{1}{2} (e^{k\tau} + e^{-k\tau}) = \cosh(k\tau),$$

and for k odd with $j' = (k+e)/2$,

$$\begin{aligned} \frac{\frac{1}{2} x^k (1 - G^{-k})}{x^{k+e} G^{-(k+e)/2}} &= \frac{1}{2} x^{-e} G^{e/2} (G^{k/2} - G^{-k/2}) \\ &= x^{-e} (1+v)^{1/2} \sinh(k\tau) = \frac{\sinh(k\tau)}{\sinh(e\tau)}, \end{aligned}$$

the last step by (6).

Set $z := \sinh(\varphi) = W^{1/2}$ and $\lambda := k/e$, so that $k\tau = \lambda\varphi = \lambda \operatorname{arcsinh}(z)$. Both $y = \cosh(\lambda \operatorname{arcsinh} z)$ and $y = \sinh(\lambda \operatorname{arcsinh} z)$ satisfy

$$(7) \quad (1+z^2) y'' + z y' - \lambda^2 y = 0.$$

Indeed for the first, $y' \sqrt{1+z^2} = \lambda \sinh(\lambda \operatorname{arcsinh} z)$; differentiating and multiplying by $\sqrt{1+z^2}$ gives $(1+z^2) y'' + z y' = \lambda^2 y$. The second is identical with $\cosh$ and $\sinh$ exchanged.

The function $\cosh(\lambda \operatorname{arcsinh} z)$ is even; write it $\sum_m a_m z^{2m}$. Reading off the coefficient of z^{2m} in (7), and using $2m(2m-1) + 2m = 4m^2$,

$$(2m+2)(2m+1)a_{m+1} + [4m^2 - \lambda^2]a_m = 0, \quad a_0 = 1,$$

so $a_{m+1} = a_m(\lambda^2 - 4m^2)/((2m+2)(2m+1))$, and telescoping with $\prod_{i=0}^{m-1} (2i+2)(2i+1) = (2m)!$ gives $a_m = \prod_{i=0}^{m-1} (\lambda^2 - 4i^2)/(2m)!$. Since $\cosh(k\tau) = \sum_m c_{k,k/2+em} W^m = \sum_m c_{k,k/2+em} z^{2m}$, substituting $\lambda = k/e$ gives (4).

The function $\sinh(\lambda \operatorname{arcsinh} z)$ is odd; write it $\sum_m b_m z^{2m+1}$. The coefficient of z^{2m+1} in (7), with $(2m+1)2m + (2m+1) = (2m+1)^2$, gives

$$(2m+3)(2m+2)b_{m+1} + [(2m+1)^2 - \lambda^2]b_m = 0, \quad b_0 = \lambda,$$

and $\prod_{i=0}^{m-1} (2i+3)(2i+2) = (2m+1)!$, so $b_m = \lambda \prod_{i=0}^{m-1} (\lambda^2 - (2i+1)^2)/(2m+1)!$. By (6), $\sinh(e\tau) = z$, so the left side in the odd case is $\sinh(\lambda \operatorname{arcsinh} z)/z = \sum_m b_m z^{2m}$, giving (5). $\square$

Two consequences are worth stating separately.

Corollary 2.3. $c_{k,j'(k)} = 1$ for k even and k/e for k odd; in both cases a 2-adic unit, e being odd. Consequently the estimate of [16, Remark 6.5] is sharp as stated: its $d(5) = 0$ is real, not the residue of a lossy bound.

Corollary 2.4. If $e \mid k$ the product in Theorem 2.2 terminates, since $\cosh(n\varphi)$ and $\sinh(n\varphi)/\sinh(\varphi)$ are Chebyshev polynomials in W . Thus $U_2(t^{-3}) = t^{-3}$ and $U_2(t^{-6}) = t^{-3} + 2t^{-6}$ are finite at $e = 3$.

Corollary 2.4 is not a curiosity. That $U_2(t^{-3}) = t^{-3}$ exactly says t^{-e} is a genuine eigenvector of eigenvalue 1, which is why the truncation parameter $\mu(P)$ cannot be taken smaller than e ; and $k = 2e$ is the first index above the truncation that still sees it. We return to this in §3.

2.3. The valuations.

Theorem 2.5 (Valuation identity). Put $\xi_i = 2i$ for k even and $\xi_i = 2i+1$ for k odd, and let s_2 denote the binary digit sum. Then

$$v_2(c_{k,j'(k)+em}) = \Sigma_m(k) - 2m + s_2(m),$$

where

$$\Sigma_m(k) := \sum_{i=0}^{m-1} [v_2(k - e\xi_i) + v_2(k + e\xi_i)].$$

Proof. By Legendre's formula $v_2(N!) = N - s_2(N)$, together with $s_2(2m) = s_2(m)$ and $s_2(2m+1) = s_2(m) + 1$,

$$v_2((2m)!) = 2m - s_2(m),$$

$$v_2((2m+1)!) = 2m+1 - (s_2(m) + 1) = 2m - s_2(m),$$

so both factorials contribute $-(2m - s_2(m))$. As e is odd, $v_2(e^{2m}) = 0$, and in the odd case $v_2(k/e) = v_2(k) = 0$. Finally the numerator factors as $\prod_i (k - e\xi_i)(k + e\xi_i)$ in both cases, since $k^2 - 4e^2i^2 = (k - 2ei)(k + 2ei)$ and $k^2 - e^2(2i+1)^2 = (k - e(2i+1))(k + e(2i+1))$. $\square$

Lemma 2.6 (Tail estimate). *For every $k \geq 1$ and every $m \geq 1$,*

$$v_2(c_{k,j'(k)+em}) \geq m.$$

Moreover $v_2 \geq m + s_2(m) \geq m + 1$ when k is odd or $4 \mid k$, and $v_2 \geq 3\lfloor m/2 \rfloor + s_2(m)$ when $k \equiv 2 \pmod{4}$. Equality $v_2 = m$ occurs only for $k \equiv 2 \pmod{4}$ and $m = 1$.

Proof. Recall e is odd.

Case k odd. Every $e\xi_i = e(2i + 1)$ is odd, so k^2 and $e^2\xi_i^2$ are both $\equiv 1 \pmod{8}$ and $v_2(k^2 - e^2\xi_i^2) \geq 3$. Hence $\Sigma_m \geq 3m$ and Theorem 2.5 gives $v_2 \geq 3m - 2m + s_2(m) = m + s_2(m) \geq m + 1$.

Case k even, say $k = 2\kappa$. Here $\xi_i = 2i$, so $k \pm e\xi_i = 2(\kappa \pm ei)$ and $\Sigma_m = 2m + \sum_{i < m} v_2(\kappa^2 - e^2i^2)$; Theorem 2.5 then gives

$$v_2(c_{k,j'(k)+em}) = \sum_{i=0}^{m-1} v_2(\kappa^2 - e^2i^2) + s_2(m).$$

If κ is even, i.e. $4 \mid k$: for i odd, ei is odd and κ even, so $\kappa \pm ei$ is odd and the term vanishes; for i even (including $i = 0$, whose term is $2v_2(\kappa) \geq 2$) both κ and ei are even and the term is ≥ 2 . There are $\lfloor m/2 \rfloor$ even i in $[0, m-1]$, so the sum is $\geq 2\lfloor m/2 \rfloor \geq m$ and $v_2 \geq m + s_2(m) \geq m + 1$.

If κ is odd, i.e. $k \equiv 2 \pmod{4}$: for i even the term vanishes; for i odd both κ and ei are odd, so the term is ≥ 3 by the mod-8 argument. There are $\lfloor m/2 \rfloor$ odd i , whence $v_2 \geq 3\lfloor m/2 \rfloor + s_2(m)$. For m even this is $\geq 3m/2 \geq m$; for m odd it is $\geq m$ if and only if $(m-3)/2 + s_2(m) \geq 0$, which holds for $m \geq 3$ since $s_2 \geq 1$, and at $m = 1$ reads $-1 + 1 = 0$.

For the equality statement: in the odd and $4 \mid k$ cases $v_2 \geq m + s_2(m) > m$. In the case $k \equiv 2 \pmod{4}$, $m = 1$ gives $v_2 = 2v_2(\kappa) + s_2(1) = 0 + 1 = 1 = m$ since κ is odd, while for $m \geq 2$ the bound $3\lfloor m/2 \rfloor + s_2(m)$ exceeds m . $\square$

2.4. What admissibility asks of a weight. A weight is a function $a : \mathbb{Z}_{\geq 1} \rightarrow \mathbb{Z}_{\geq 0}$; it enters through the growth module

$$(8) \quad \mathcal{A}_{\pi,P}^{\mathbf{m},*} := \left\{ \sum_k b_k t_P^{-k} : v_\pi(b_k) \geq a(k)/m_P \text{ for } k > 0 \right\},$$

and on the side of [14] through the coefficientwise module $\mathcal{D} = \prod_{n \in \mathbb{Z}} p^{b(n)} t^n \mathcal{O}_L$ with $b(-k) = a(k)$ (§5.1). Define the *defect*

$$(9) \quad d(k) := \min_{m \geq 0} [a(k) - a(j'(k) + em) + v_2(c_{k,j'(k)+em})].$$

The conditions the global argument consumes are the following, each with its consumer named. They are read off [16, §§6.1.2, 6.2, Corollaries 6.7, 6.8, Lemma 7.11] and, on the other side, [14, §7.2, Lemma 6.12, eqs. (21), (22)].

- (A1) $a(k) = 0$ for $k \leq \mu(P) = e$. Consumed by the exact sequence of [16, §6.2] — the kernel is the global sections of a line bundle, so poles of order $\leq \mu(P)$ must be unweighted — and by [14, eq. (21)].
- (A2) $a(k) = O(k)$. Consumed by [16, Corollary 6.7], which needs $\bigcup_m \tilde{V}^m = \tilde{V}^\dagger$.

- (A3) $d(k) \geq 1$ for $k > \mu(P)$, and $d(k) \rightarrow \infty$. The bound ≥ 1 has a unique consumer, [16, Lemma 7.11], and on the other side [14, §7.2, case (II)]; the divergence is what makes the operator nuclear.
- (A4) $a(k) \geq d(k)$. Consumed in the proof of [16, Proposition 6.6(2)] and by the “ $b(n) \geq m$ ” step of [14, §7.2, case (II)].
- (A5) $d(k) \geq 0$ for $k \leq \mu(P)$. This is $\tilde{\Theta}$ -stability in [16, Corollary 6.8] and $U_p \circ C(\mathcal{O}_{\mathcal{R}}^{\text{con}}) \subset \mathcal{O}_{\mathcal{R}}^{\text{con}}$ in [14, eq. (22)].

Two remarks on the calibration. First, $v_\pi(c) = v_\pi(p)v_2(c)$ and $m_P \geq 1/v_\pi(p)$, so the v_2 term enters (9) with coefficient at least 1; using it with coefficient exactly 1 is therefore conservative on the route of [16], and legitimate because $v_2(c_{k,j}) \geq 0$ — which is now a consequence of Lemma 2.6 for $m \geq 1$ and Corollary 2.3 for $m = 0$, rather than an assumption. On the route of [14] there is no growth parameter and no π at all, so the requirement is literally $d(k) \geq 1$ in v_p and the calibration is exact.

Second, since the $m = 0$ term of (9) equals $a(k) - a(j'(k))$ by Corollary 2.3, condition (A3) splits into

- (A3a) $a(k) - a(j'(k)) \geq 1$ for $k > \mu(P)$, tending to infinity;
- (A3b) $a(j'(k) + em) - a(j'(k)) \leq v_2(c_{k,j'(k)+em})$ for $m \geq 1$,

the second saying that no tail term beats the leading one, so that $d(k) = a(k) - a(j'(k))$.

Remark 2.7 (No admissible weight makes the module a ring, at any p). Closure of (8) under multiplication would need $a(i + j) \leq a(i) + a(j)$; with (A1)’s $a(e) = 0$ that forces $a(k + e) \leq a(k)$, contradicting the divergence in (A3). The module B^m of [16, Definition 6.3] is not a ring either, u^{-1} violating its own growth condition. This matters in §5: the exactness statement that Theorem 1.2 needs is a pre-existing feature of the source, not a cost introduced by changing the weight.

2.5. The repaired weight.

Theorem 2.8 (Admissibility). *Let $p = 2$, $e = 3$, $\mu(P) = 3$, and*

$$(10) \quad a(k) = \begin{cases} 0, & k \leq 3, \\ \left\lfloor \frac{k-1}{3} \right\rfloor + (k \bmod 2), & k \geq 4. \end{cases}$$

Then (A1)–(A5) hold; $d(k) \geq 1$ for every $k > 3$; the minimum in (9) is attained at $m = 0$, so that $d(k) = a(k) - a(j'(k))$; and $d(k) \geq 2\lfloor k/12 \rfloor - 1 \rightarrow \infty$, with $d(k) \sim k/6$.

The weight is *integer-valued*, which is not cosmetic: on the route of [14] it appears as $p^{a(k)}$, which then lies in $\mathcal{O}_L$ with no base extension. The pointwise-extremal weight attaining the same rate takes values in $\frac{1}{6}\mathbb{Z}$ and would require one. It also differs from Kramer–Miller–Upton’s own weight by at most 1 at every index, so nothing that depends on the weight’s growth class changes.

Proof. (A1) holds by definition, and (A2) because $a(k) \leq (k+2)/3$. For (A5), $a(k) = 0$ for $k \leq 3$ gives $d(k) = 0 \geq 0$ there. (A4) follows once (A3) is known: by Corollary 2.3, $d(k) \leq a(k) - a(j'(k)) \leq a(k)$ since $a \geq 0$.

The increment formula. For $n \geq 4$ and $m \geq 1$ one has $\lfloor (n+3m-1)/3 \rfloor = \lfloor (n-1)/3 \rfloor + m$ and $(n+3m) \bmod 2 = (n+m) \bmod 2$, so

$$(11) \quad a(n+3m) - a(n) = \begin{cases} m, & m \text{ even}, \\ m+1, & m \text{ odd, } n \text{ even}, \\ m-1, & m \text{ odd, } n \text{ odd}. \end{cases}$$

It also holds at $n = 2$, where $a(2) = 0 = \lfloor 1/3 \rfloor + 0$, and *fails* at $n = 1$ and $n = 3$, where (A1) overrides the closed form. Both exceptions are harmless: for $k > 3$ the value $j'(k) = 1$ never occurs (k even gives $j' \geq 2$, k odd gives $j' \geq 4$), and $j'(k) = 3$ occurs only for $k = 6$.

Proof of (A3b). Let $k > 3$, $j' = j'(k)$, $m \geq 1$; we must show $a(j' + 3m) - a(j') \leq v_2(c_{k,j'+3m})$.

- $j' = 3$. Then $k = 2j' = 6$, the odd alternative $k = 2j' - 3 = 3$ being excluded by $k > \mu(P) = 3$. By Theorem 2.2 with $\lambda = 2$ the product has the factor $4 - 4 = 0$ at $i = 1$, so $c_{6,j'+3m} = 0$ for $m \geq 2$ and only $m = 1$ constrains: $a(6) - a(3) = 1$ and $c_{6,6} = \lambda^2/2! = 2$, so $v_2 = 1$. Equality.
- m even. The increment is m and Lemma 2.6 gives $v_2 \geq m$.
- m odd, j' odd. The increment is $m - 1 < m \leq v_2$.
- m odd, j' even. The increment is $m + 1$. The indices k with $j'(k) = j'$ are $k = 2j'$ and $k = 2j' - 3$; j' even makes $2j' \equiv 0 \pmod{4}$, and $2j' - 3$ is odd, so in *both* cases the refinement in Lemma 2.6 gives $v_2 \geq m + s_2(m) \geq m + 1$.

Observe why the parity indicator is exactly the right correction: Lemma 2.6 is tight only at $k \equiv 2 \pmod{4}$, $m = 1$, that is at $j' = k/2$ *odd* — precisely the case in which the indicator makes the increment go *down* by one rather than up. The two tightnesses are complementary, which is why $\lfloor (k-1)/3 \rfloor$ alone fails at $k = 5$ and $\lfloor (k-1)/3 \rfloor + (k \bmod 2)$ does not.

Proof of (A3a). Write $c := j'(k)$, so $k = 2c$ for k even and $k = 2c - 3$ for k odd, in which case $c \geq 4$ because $k > 3$. Put $c = 6\mathfrak{q} + r$ with $0 \leq r \leq 5$ and use $\lfloor (6\mathfrak{q} + s)/3 \rfloor = 2\mathfrak{q} + \lfloor s/3 \rfloor$.

For $k = 2c$ even, $a(2c) = \lfloor (12\mathfrak{q} + 2r - 1)/3 \rfloor = 4\mathfrak{q} + \lfloor (2r - 1)/3 \rfloor$ while $a(c) = 2\mathfrak{q} + \lfloor (r - 1)/3 \rfloor + (r \bmod 2)$, so $d = 2\mathfrak{q} + \lfloor (2r - 1)/3 \rfloor - \lfloor (r - 1)/3 \rfloor - (r \bmod 2)$. For $k = 2c - 3$ odd, $a(k) = \lfloor (2c - 4)/3 \rfloor + 1 = 4\mathfrak{q} + \lfloor (2r - 4)/3 \rfloor + 1$, so $d = 2\mathfrak{q} + \lfloor (2r - 4)/3 \rfloor - \lfloor (r - 1)/3 \rfloor - (r \bmod 2) + 1$. The two evaluate identically:

r	0	1	2	3	4	5
d	$2\mathfrak{q}$	$2\mathfrak{q} - 1$	$2\mathfrak{q} + 1$	$2\mathfrak{q}$	$2\mathfrak{q} + 1$	$2\mathfrak{q} + 1$

For $r \in \{2, 4, 5\}$ we get $d \geq 1$ for every $\mathfrak{q} \geq 0$. For $r \in \{0, 1, 3\}$ we need $\mathfrak{q} \geq 1$, and the cases $\mathfrak{q} = 0$ are $c \in \{0, 1, 3\}$: on the even branch $c = 0, 1$ give $k = 0, 2$, excluded by $k > 3$, and $c = 3$ gives $k = 6$, where the closed form for

$a(c)$ is invalid and (A1) applies, $d(6) = a(6) - a(3) = 1$; on the odd branch $c \geq 4$, so none occurs. Finally $c = 2$ (that is $r = 2$, $\mathfrak{q} = 0$, $k = 4$) uses the validity of (11) at $n = 2$: $d(4) = a(4) - a(2) = 1$.

Divergence. In both branches $d \geq 2\mathfrak{q} - 1 = 2\lfloor c/6 \rfloor - 1$, and $c = j'(k) \geq k/2$, so $d(k) \geq 2\lfloor k/12 \rfloor - 1 \rightarrow \infty$; asymptotically $d(k) \sim c/3 \sim k/6$. $\square$

Remark 2.9 (Machine confirmation). An exact-rational sweep over $4 \leq k \leq 400$ and the full support $m \leq 259$ finds no violation of (A3), the minimum attained at $m = 0$ for every k , and $\min d = 1$, with $d(100) = 17$, $d(200) = 33$ and $d(400) = 67$; the control run with $a(k) = \lfloor (k-1)/3 \rfloor$ reproduces $d(5) = 0$. The coefficients are computed twice, once from Theorem 2.2 and once by a from-scratch series solve of (3) that never uses it, agreeing on all 355 pairs examined for $e \in \{1, 3, 5, 7\}$; Theorem 2.5 is checked to $k \leq 60$ and Lemma 2.6 to $k \leq 600$, $m \leq 80$, where the 150 tight pairs found are all of the predicted shape.

3. SHARPNESS

The rate $k/6$ delivered by Theorem 2.8 is not an artifact of the particular weight (10). It is the best rate any weight can achieve, and the obstruction is a single coefficient.

Theorem 3.1 (Sharpness). *Let $p = 2$ and let e be odd. Then $k = 2e$ is a self-loop of the support map — that is, $2e \in j'(2e) + e\mathbb{Z}_{\geq 0}$ — and*

$$c_{2e, 2e} = 2, \quad \text{hence} \quad d(2e) \leq v_2(2) = 1$$

for every weight whatsoever. Consequently, at $e = 3$, a target $d(k) \geq \max(1, \gamma k)$ is achievable if and only if $\gamma \leq 1/6$, and $d(k) \geq \max(1, k/6)$ is achieved by (10). At general odd e the threshold is $\gamma \leq 1/(2e)$.

Proof. $k = 2e$ is even, so $j'(2e) = e$ by Lemma 2.1, and $j'(2e) + e = 2e = k$: the index k occurs in its own support, at $m = 1$. Theorem 2.2 with $\lambda = k/e = 2$ gives

$$c_{2e, e+e} = \frac{\lambda^2 - 0}{2!} = \frac{4}{2} = 2,$$

independently of e — the substitution $\lambda = 2$ erases e entirely. The constraint (A3b) at $(k, j) = (2e, 2e)$ therefore reads

$$d(2e) \leq a(2e) - a(2e) + v_2(c_{2e, 2e}) = 1,$$

in which the weight cancels identically. Since $d(2e) \leq 1$ while a target γk at $k = 2e$ would demand $d(2e) \geq 2\gamma e$, we need $\gamma \leq 1/(2e)$; at $e = 3$ that is $\gamma \leq 1/6$, attained by Theorem 2.8. $\square$

The same mechanism recurs. The index $k = 4e$ is even with $j' = 2e$ and $j' + 2e = k$, so it is a self-loop at $m = 2$; Theorem 2.2 with $\lambda = 4$ gives $c_{4e, 4e} = 16 \cdot 12/4! = 8$, again independent of e . Both values are confirmed by exact computation for $e = 1, 3, 5, 7, 9, 11, 13, 15, 21$ and $e = 3, 5, 7, 9, 11$ respectively.

Remark 3.2 (What Remark 6.5 is really about). The descending orbit of the successor map $k \mapsto j'(k)$ has exactly two attractors below $\mu(P) = e = 3$: the fixed point 3 and the 2-cycle $\{1, 2\}$. By Corollary 2.4, $U_2(t^{-3}) = t^{-3}$ is a genuine eigenvector of eigenvalue 1, which is why the truncation $\mu(P) = e$ is forced rather than chosen; and $k = 6$ is the first index above the truncation that still sees it. The obstruction *is* that eigenvector, seen one step up.

3.1. Consequence for the contact theory. Theorem 3.1 has a consequence for Theorem 1.2 that is worth isolating, because it says the restriction there cannot be traded away.

The contact criterion of [16, Theorem 1.1] is stated for a q -adic truncation parameter $r \in [0, 1]$. Its proof runs through [16, Definition 7.3(2)], which asks $d(k)/M \geq r$ for every $k > \mu(P)$, where $M = m_{e,P}$ is the growth parameter of the local module. The reduction that makes the sequence of [16, §6.2] available for the repaired weight needs $M \geq 1$. Combining that with Theorem 3.1 gives π -adic $r \leq 1$ whatever M is:

Corollary 3.3. *No choice of auxiliary tame index e relieves the restriction in Theorem 1.2. For every odd e and every weight, $d(2e) \leq 1$; with $M \geq 1$ this caps the π -adic truncation parameter at 1 regardless of the normalisation.*

The conversion to the q -adic parameter of [16, Theorem 1.1] is where the order of ρ enters. Since $\pi_q = \pi^{v_p(q)}$ we have $v_{\pi_q}(x) = v_\pi(p) v_q(x)$; at $p = 2$ with ρ of order 2^n the ring $R = \mathbb{Z}_2[\zeta_{2^n}]$ is totally ramified of degree 2^{n-1} , so $v_\pi(p) = 2^{n-1}$ and π -adic $r \leq 1$ covers exactly q -adic $r \leq 2^{1-n}$. Three independent readings of the source agree on this factor: the glossary definition of π_q ; the ratio $d_P(p-1)/\delta_P = v_\pi(p)$ between the two printings of the same local polygon; and the window count $\deg L(\rho_P^{\text{ext}}, s) = d_P - 1$. At order 2 the factor is 1 and the full range of [16, Theorem 1.1] is covered.

There is a genuine tension here, and we state it as such rather than resolving it: [16] *enlarge e* exactly where the reduction shrinks it. The proof of their Theorem 7.13 reads “since $r \leq v_\pi(p)$, we may enlarge e as necessary and assume that $r \leq e$ ”. Theorem 3.1 says that enlarging e buys nothing at the index $2e$.

Remark 3.4 (Open problem). Two routes would remove the restriction in Theorem 1.2, and neither has been taken. The first is to prove the exact sequence of [16, §6.2] outright; it is asserted without proof there, at every p and for their own weight, only the $A^\dagger$ version (their Lemma 5.15) being proved. By Remark 2.7 this is not a cost introduced by changing the weight: no admissible weight makes the module a ring, at any p . The second is to re-run [16, §6.2] and the whole of their §7 in the coefficientwise formulation of [14] — which is available in principle, and is exactly why Theorem 1.1 carries no cap — but their perturbation machinery is built on the B^{m_e} basis and the tuple m_e , so transporting it is a research problem rather than a citation.

A third, partially explored option: compare $\mathcal{A}_{\pi, P}^{\mathbf{m},*}$ (weight a , parameter m_P) against the module of [16] with their weight and parameter $m_P/2$. Since $a \leq 2a_{\text{KMU}}$ for $k \geq 4$ this gives a containment, and would relax the cap to q -adic $r \leq 1/2$ uniformly in n ; whether the rest of the argument survives the substitution has not been checked.

4. THE CHARACTERISTIC-2 TAME BELYI MAP

4.1. What degenerates at $p = 2$. The map the global argument needs is described in [16, §4.1] as a composition

$$\eta_q : \mathbb{P}^1 \xrightarrow{(q-1)\text{-power}} \mathbb{P}^1 \xrightarrow{\text{linear}} \mathbb{P}^1 \xrightarrow{(p-1)\text{-power}} \mathbb{P}^1 \xrightarrow{\text{linear}} \mathbb{P}^1,$$

the second map fixing 1 and ∞ and sending 0 to another $\mathbb{F}_q$ -rational point, the last swapping 0 and 1. Precomposed with a tame map $\eta_0 : X \rightarrow \mathbb{P}^1$ this yields [16, Proposition 4.3]: a tame Belyi map η with $\eta(P) = 0$ for each $P \in S$, and ramification index $p - 1$ at every P with $\eta(P) = 1$.

At $p = 2$ this degenerates in three independent ways, all of them the single fact $\mu_{p-1} = \{1\}$: the map $z \mapsto z^{p-1}$ is the identity, so no ramification is created; the resulting index $e_P = 1$ is incompatible with the degree identity below whenever $g \geq 1$; and the auxiliary point that [15, Lemma 3.1] places at $c = 2$ is the point 0 in characteristic 2, so even the placement step is empty. In addition the input is unavailable: η_0 is obtained at $p \geq 3$ from Fulton's simply-branched map [9], and simple branching is not tame at $p = 2$.

Both halves are replaceable. The tame map exists over a finite field at every characteristic by [13, Theorems 1.2 and 7.6], resting on [23, Theorem 1.1] over an algebraically closed field; and *tame* is all that [16, Theorem 4.1] asks of it. The $(p - 1)$ -power stage is replaced by an e -power stage for any odd $e \mid q - 1$.

4.2. The construction.

Lemma 4.1 (Tame Belyi map of uniform index e). *Let $q = 2^a$, let X be a smooth projective geometrically irreducible curve over $\mathbb{F}_q$, let $S \subset X$ be a finite set of closed points, and let $e > 1$ be odd with $e \mid q - 1$. Then, after enlarging $\mathbb{F}_q$ if necessary — only to make branch points and S rational and q large — there is a tame Belyi map $\eta : X \rightarrow \mathbb{P}_{\mathbb{F}_q}^1$, finite and separable, ramified only over $\{0, 1, \infty\}$ with all branch points $\mathbb{F}_q$ -rational, such that*

- (1) $\eta(P) = 0$ for every $P \in S$;
- (2) every $P \in \eta^{-1}(1)$ has ramification index exactly e ;
- (3) hence $r_1 e = \deg \eta$, $\eta^*(1) = \sum_{\eta(P)=1} eP$, and

$$(12) \quad 2(g - 1) + r_0 + r_1 + r_\infty = \deg \eta,$$

where $r_Q = |\eta^{-1}(Q)|$.

Proof. Step 0 (the tame map). By [13, Theorem 1.2] there is a finite separable tame $\eta_0 : X \rightarrow \mathbb{P}_{\mathbb{F}_q}^1$; no base extension is needed for this step. Enlarge $\mathbb{F}_q$ so that every branch point of η_0 and every point of S is rational, that $\mu_e \subset \mathbb{F}_q$

(automatic from $e \mid q - 1$), and that $q + 1 > |\text{Branch}(\eta_0) \cup \eta_0(S)| + 2$; then choose the coordinate on the target so that 0 and ∞ avoid $\text{Branch}(\eta_0) \cup \eta_0(S)$, which therefore lies in $\mathbb{F}_q^\times$.

Step 1 (the $(q - 1)$ -power map). Let $g_1(z) = z^{q-1}$. Since $q - 1$ is odd, g_1 is tame; its derivative $(q - 1)z^{q-2} = z^{q-2}$ vanishes only at 0, so it is totally ramified over 0 and ∞ with index $q - 1$ and unramified elsewhere, and $g_1(\mathbb{F}_q^\times) = \{1\}$. Put $\phi := g_1 \circ \eta_0$. Then $\text{Branch}(\phi) \subset \{0, 1, \infty\}$ and $\phi(S) = \{1\}$; and because 0 is not a branch point of η_0 , every point of $\phi^{-1}(0) = \eta_0^{-1}(0)$ has index exactly $q - 1$.

Step 2 (the linear map). Choose $c \in \mu_e(\mathbb{F}_q) \setminus \{1\}$, nonempty since $e \mid q - 1$ and $e > 1$, and let $g_2 \in \text{PGL}_2$ fix 1 and ∞ with $g_2(0) = c$. Then $\text{Branch}(g_2\phi) = \{c, 1, \infty\}$ and $(g_2\phi)(S) = \{1\}$, and the point 0 is *not* in this branch locus, c being a root of unity.

The role of g_2 is not decorative. Its sole job is to evacuate the point 0 from the branch locus, so that the fibre over 0 is clean for Step 3. Without it, $0 \in \text{Branch}(\phi)$ and the points of $(g_3\phi)^{-1}(0)$ would have index $e \cdot e_\phi(\cdot)$ with e_ϕ varying over the fibre; the fibre over the final 1 would then have *mixed* index and claim (2) would be false.

Step 3 (the e -power map). Let $g_3(z) = z^e$. Since $\gcd(e, 2) = 1$, g_3 is tame; it is totally ramified over 0 and ∞ with index e , unramified elsewhere, and $g_3^{-1}(1) = \mu_e \ni c, 1$. Hence, using $c^e = 1$,

$$\text{Branch}(g_3g_2\phi) \subset g_3(\{c, 1, \infty\}) \cup \{0, \infty\} = \{0, 1, \infty\}.$$

The fibre $g_3^{-1}(0) = \{0\}$ has index e , and 0 is unramified for $g_2\phi$ by Step 2, so every point of $(g_3g_2\phi)^{-1}(0)$ has index exactly e ; there are $\deg(g_2\phi) = (q - 1) \deg \eta_0$ of them.

Step 4 (the swap). Let $g_4 \in \text{PGL}_2$ fix ∞ and swap 0 and 1; it carries the S -fibre, over 1, to 0, and the uniform-index- e fibre, over 0, to 1. Set $\eta := g_4g_3g_2g_1\eta_0$. Claims (1) and (2) hold.

Tameness. Ramification indices multiply and every factor is odd: e_{η_0} is odd because η_0 is tame at $p = 2$; $q - 1$ is odd; e is odd; linear maps are unramified. Explicitly $e_\eta(P) = e_{\eta_0}(P)$ for $P \in S$, $e_\eta(P) = e$ for P over 1, and $e_\eta(P) = e(q - 1)e_{\eta_0}(P)$ for P over ∞ . So η is tame, and finite and separable as a composite of such.

Claim (3). All points over 1 have index e and indices over a point sum to the degree, so $r_1e = \deg \eta$ and $\eta^*(1) = \sum_{\eta(P)=1} eP$. Tame Riemann–Hurwitz for η , unramified outside $\{0, 1, \infty\}$, gives

$$2g - 2 = -2 \deg \eta + \sum_{Q \in \{0, 1, \infty\}} \sum_{P|Q} (e_P - 1) = -2 \deg \eta + \sum_Q (\deg \eta - r_Q),$$

which is (12). □

Two features of the construction should be stated explicitly, since both are things a reader checks first.

Remark 4.2 (The fibre over the final 0 is a genuine mixture). It contains the points of S with index e_{η_0} , the points of $\eta_0^{-1}(0)$ arriving via c with index $q - 1$, and $e - 2$ unramified sheets from the remaining e -th roots of unity. This is harmless: [16] asks only $\eta(S) = \{0\}$, and [14] only that the points at infinity lie in $\eta^{-1}(\{0, \infty\})$. Neither asks that the whole fibre be S .

Remark 4.3 (Where the second ramification point of $z \mapsto z^e$ goes). The map $z \mapsto z^e$ ramifies at 0 and at ∞ . The point 0 is the wanted one and becomes the fibre over the final 1; the point ∞ maps to ∞ , which g_4 fixes, so the second ramification point lands over ∞ with index $e(q - 1)e_{\eta_0}$ — large, but odd, hence tame. It creates no new type-2 point and does not meet S . The reason is that at $\eta(Q) \in \{0, \infty\}$ the local Frobenius is a *pure* p -th power: [14, §3.4] takes the local parameter to be $u_Q = t^{\pm 1/e_Q}$ with a general e_Q and records $\sigma(u_Q) = u_Q^q$, so the unique e_Q -th root congruent to t^p modulo p is t^p itself whatever e_Q is. Only over 1, where $\sigma(u) = (u + 1)^p - 1$ is not a pure power, does the root extraction produce the factor G of §2.1.

4.3. The auxiliary point must be a root of unity.

Proposition 4.4. *In the construction of §4.1, the target c of the linear map must satisfy $c^{p-1} = 1$. The description in [16, Proposition 4.3] of that map as sending 0 to “any other $\mathbb{F}_q$ -rational point” over-states the freedom, at every p .*

Proof. After the linear map the branch locus is $\{c, 1, \infty\}$; after $z \mapsto z^{p-1}$ it is $\{c^{p-1}, 1, \infty\} \cup \{0, \infty\}$. For the composite to be Belyi one needs $c^{p-1} \in \{0, 1, \infty\}$, and $c \notin \{0, \infty\}$ excludes the first and third, forcing $c^{p-1} = 1$, that is $c \in \mu_{p-1}(\mathbb{F}_q) \setminus \{1\}$. $\square$

This is load-bearing rather than cosmetic: with a generic c the composite acquires a *fourth* branch point, $\eta^{-1}(V)$ is not étale over V , and the lifting of [16, §4.2] has no basis. But no published mathematics is wrong. The original, [15, Lemma 3.1], puts the branch points at $\{1, 2, \infty\}$, that is $c = 2$, and $2^{p-1} = 1$ in $\mathbb{F}_p$ by Fermat with $2 \neq 1$ for $p \geq 3$. The defect is confined to the paraphrase in [16], and the repair is one word: $c \in \mu_{p-1}(\mathbb{F}_q) \setminus \{1\}$, a set that is nonempty *exactly because* $p \geq 3$. At $p = 2$ the point “2” is the point 0, which is the third degeneration of §4.1.

4.4. Degree three forces $3 \mid q - 1$.

Proposition 4.5. *A degree-3 auxiliary map with the properties Steps 2–4 of Lemma 4.1 exists over $\mathbb{F}_q$, $q = 2^a$, if and only if $3 \mid q - 1$, that is if and only if a is even.*

Proof. Write the auxiliary stage as one map $h = g_4 g_3 g_2 : \mathbb{P}^1 \rightarrow \mathbb{P}^1$. It is required to be tame, with $\text{Branch}(h) \subset \{0, 1, \infty\}$ and $h(\{0, 1, \infty\}) \subset \{0, 1, \infty\}$ and $h(1) = 0$, and with every $y \in h^{-1}(1)$ satisfying $e_h(y) = 3$ and $y \notin \{0, 1, \infty\}$ — the last so that the earlier stage, whose branch locus is $\{0, 1, \infty\}$, is unramified there. Sufficiency is Lemma 4.1 with $e = 3$.

For necessity, Riemann–Hurwitz for a tame degree-3 self-map of $\mathbb{P}^1$ in characteristic 2 gives $\sum_P (e_P - 1) = 4$ with every e_P odd — $e_P = 2$ being wild — and $e_P \leq 3$, so $e_P \in \{1, 3\}$ and there are exactly two totally ramified points. Call them α , lying over 1, and β ; then $h^*(1) = 3[\alpha]$ and $h^*(w) = 3[\beta]$ with $w := h(\beta) \in \{0, \infty\}$, since the fibre over 1 is $\{\alpha\}$, and h is unramified elsewhere. Four configurations remain, and each forces $\alpha^3 = 1$ with $\alpha \neq 1$; we work in characteristic 2 throughout.

$w = \infty, \beta = \infty$. Then h is a cubic polynomial; $h(0) = \infty$ is impossible and $h(0) = 1$ would force $\alpha = 0$, so $h(0) = 0$. Total ramification at α over 1 gives $h = \lambda(z + \alpha)^3 + 1$, so $h(1) = 0$ gives $\lambda(1 + \alpha)^3 = 1$ and $h(0) = 0$ gives $\lambda\alpha^3 = 1$, whence $(1 + \alpha^{-1})^3 = 1$. Setting $\gamma := 1 + \alpha^{-1} \neq 1$, γ is a primitive cube root of unity, and $\gamma^2 + \gamma + 1 = 0$ gives $\alpha = \gamma^{-2} = \gamma$.

$w = \infty, \beta \neq \infty$. Then $h(\infty) = 0$. If $\beta = 0$ then $h = \lambda(z + \alpha)^3/z^3$ and $h(1) = 0$ forces $\alpha = 1$, excluded. So $\beta \notin \{0, 1, \infty\}$, $h^*(0) = [0] + [1] + [\infty]$ and $h = \lambda z(z + 1)/(z + \beta)^3$. Matching $(z + \beta)^3 + \lambda z(z + 1) = (z + \alpha)^3$ gives $\beta + \lambda = \alpha$, $\beta^2 + \lambda = \alpha^2$ and $\beta^3 = \alpha^3$. The first two give $(\alpha + \beta)^2 = \alpha + \beta$, so $\beta = \alpha$ — whence $\lambda = 0$, degenerate — or $\beta = \alpha + 1$; the third then gives $(\alpha + 1)^3 = \alpha^3$, that is $\alpha^2 + \alpha + 1 = 0$.

$w = 0$. Then $h^*(0) = 3[\beta]$, and $h(1) = 0$ forces $\beta = 1$; also $h(0) = h(\infty) = \infty$, the alternatives forcing $0 = 1$ or $\alpha \in \{0, \infty\}$. So $h = \lambda(z + 1)^3/(z(z + \beta'))$ for a third simple pole β' , and matching $\lambda(z + 1)^3 + z(z + \beta') = \lambda(z + \alpha)^3$ gives, from the constant coefficient, $\lambda = \lambda\alpha^3$, that is $\alpha^3 = 1$; and $\alpha = 1$ would make the z^2 coefficient read $\lambda + 1 = \lambda$. The remaining configuration, $w = 0$ with $\beta \neq 1$, is impossible since $h(1) = 0$ and the fibre over 0 is $\{\beta\}$.

So α is a primitive cube root of unity in $\mathbb{F}_q$, whence $3 \mid q - 1$. $\square$

An independent enumeration over $\overline{\mathbb{F}}_q$ for $a \leq 6$ confirms the pattern and the set of admissible α : no such map for $q = 2, 8, 32$, and for $q = 4, 16, 64$ exactly the maps with $\alpha \in \mu_3(\mathbb{F}_q) \setminus \{1\}$.

Remark 4.6 (Explicit instance). For $X = \mathbb{P}^1$, $\eta_0 = \text{id}$, $S \subset \mathbb{F}_q^\times$ and ω a primitive cube root of unity, Lemma 4.1 produces $\eta(z) = ((1 + \omega)z^{q-1} + \omega)^3 + 1$ of degree $3(q - 1)$, with $r_1 = q - 1$ points of index exactly 3 over 1, $r_0 = 2q - 1$ and $r_\infty = 1$. Then (12) reads $2(0 - 1) + (2q - 1) + (q - 1) + 1 = 3q - 3$ and $\sum_P (e_P - 1) = 6q - 8 = 2 \deg \eta - 2$, so Riemann–Hurwitz is saturated by these three fibres and η really is Belyi. Verified symbolically and numerically over $\mathbb{F}_4$ and $\mathbb{F}_{16}$ in the replication package.

4.5. The base change is free. Proposition 4.5 makes η at $e = 3$ available only when a is even. For a odd one passes to $\mathbb{F}_{q^2}$, and this costs nothing. Both source papers already license unspecified extensions — [14, §2.1] says “it is enough to prove Theorem 1.1 after replacing q with a larger power of p ” — but neither proves the reduction, and here it carries weight, so we prove it.

Proposition 4.7 (Base-change invariance). *Let $\mathbb{F}_{q^m}/\mathbb{F}_q$ be finite and $\rho' = \rho|_{\pi_1(X_{\mathbb{F}_{q^m}})}$, and assume ρ' nontrivial. Then $\text{NP}_{q^m}(\rho') = \text{NP}_q(\rho)$ and*

$\text{HP}_{q^m}(\rho') = \text{HP}_q(\rho)$ as polygons. Hence Theorem 1.1 over $\mathbb{F}_{q^m}$ implies it over $\mathbb{F}_q$.

Proof. Newton side. The cohomology $H_c^1(X_{\overline{\mathbb{F}}_q}, \mathcal{F}_\rho)$ is geometric, hence unchanged, and $\text{Frob}_{q^m} = \text{Frob}_q^m$, so the inverse roots become α_i^m . In the q^m -adic normalisation

$$v_{q^m}(\alpha_i^m) = \frac{v_p(\alpha_i^m)}{v_p(q^m)} = \frac{m v_p(\alpha_i)}{m v_p(q)} = v_q(\alpha_i),$$

so the multiset of normalised valuations, and hence the polygon, is unchanged.

Hodge side. By (1) the polygon has length $2(g - 1 + |S|) + \sum_P (d_P - 1)$. If $|S|$ counted closed points of $X/\mathbb{F}_q$ then base change would change it — a closed point of degree d splits into $\gcd(d, m)$ points — and HP would *not* be invariant. The degree count forces the geometric reading:

$$2(g - 1 + |S|) + \sum_P (d_P - 1) = 2g - 2 + |S| + \sum_P d_P = -\chi_c(X, \mathcal{F}_\rho)$$

by Grothendieck–Ogg–Shafarevich for a rank-1 sheaf, and that formula sums over *geometric* points. So $|S|$ is the geometric count and the d_P are geometric Swan conductors; g is geometric; and Swan conductors are invariant under unramified base change. Every ingredient of the slope multiset is unchanged, as is ordinarity of X . $\square$

The nontriviality hypothesis is not vacuous — ρ' can become trivial for a character pulled back from $\text{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ — but it cannot fail here, because ρ is wildly ramified at some point of S and ramification is geometric.

Remark 4.8 (An extension-free variant, and why it is not a drop-in). Deleting Steps 2 and 3 of Lemma 4.1 and setting $\eta := g_4 \circ g_1 \circ \eta_0$ also works: by Step 1 every point of $\phi^{-1}(0)$ already has index exactly $q-1$, odd and > 1 for $q \geq 4$, so g_4 alone gives a tame Belyi map with $\eta(S) = \{0\}$ and uniform index $e = q-1$ over 1, with no root-of-unity condition and no auxiliary stage. More generally the e -power stage exists exactly when $e \mid q-1$, which is *simultaneously* the condition for $\mu_e \subset \mathbb{Z}_q$, that is for the eigenspace decomposition of [16, Definition 6.3]; at odd p one has $p-1 \mid q-1$ automatically, which is why $p-1$ is their choice.

This variant is insurance, not a route. It moves $\mu(P)$ to $q-1$, so (A1) becomes $a(k) = 0$ for $k \leq q-1$; and while Theorems 2.2 and 2.5, Lemma 2.6 and Theorem 3.1 hold for *every* odd e , the mod-6 case analysis of Theorem 2.8 is specific to $e = 3$ and would have to be redone modulo $2e$ for a field-dependent e , once per q . That is a research task rather than a substitution. We take the base extension instead, which Proposition 4.7 makes free.

5. ASSEMBLY

Theorem 2.8 is stated in the language of [16], and the global argument that carries Theorem 1.1 is [14]’s. This section closes that gap, records what the

global argument consumes and why each item is p -uniform, and assembles the proof.

5.1. The dictionary. Both papers control the same operator on the same kind of module; they write it differently. Writing ν for the p -Frobenius of [14, §4.2] — which is the map §2.1 calls σ , that paper reserving σ for the a -th iterate — and b for its weight:

[14]	[16] and here	agree?
$\nu(t) = ((t^{p-1} + 1)^p - 1)^{1/(p-1)}$	the type-2 σ of §2.1	identical
$b(-K) = \lfloor (K - 1)/(p - 1) \rfloor$	$a_{\text{KMU}}(K)$, same formula	identical under $n = -K$
$K = k + np$, $0 \leq k \leq p - 1$, gain p^n	$K = p\ell + r$, $0 \leq r < p$, gain π^{ℓ/m_P}	identical
$\mathcal{D} = \prod_n p^{b(n)} t^n \mathcal{O}_L$	$\mathcal{A}_{\pi, P}^{\mathbf{m},*}$ of (8)	same shape; $\mathcal{D}$ coefficientwise
$U_p(\mathcal{D}) \subset \mathcal{D}$	(A5), and (A3) in weak form	identical

The entry that matters is the fourth. The module $\mathcal{D}$ of [14, §4.2] is defined *coefficientwise*, by integer exponents, with no radius parameter and no π anywhere. Two consequences follow, and both are used above. First, a weight that is not a Galois-eigenspace regrading — which (10) is not — is native there, whereas [16, Definition 6.3] builds its module as an eigenspace sum. Second, the calibration is exact rather than conservative: the requirement really is $d(k) \geq 1$ in v_p , on the nose (§2.4). Hence Lemma 2.6 and Theorem 2.8 transport verbatim: same operator, same weight function under $n \leftrightarrow -K$, same conclusion.

5.2. What the global argument consumes. We substitute into [14] at exactly two places: its Lemma 3.1, the geometric input, is replaced by Lemma 4.1; and its §4.2 with Proposition 4.2, the type-2 local estimate, is replaced by Theorems 2.2, 2.5 and 2.8 and Lemma 2.6 at $e = 3$ with the weight (10). Everything else is cited. The following are the load-bearing items, each with the reason it does not see the prime.

(C1) *The truncation and the Riemann–Roch count.* The truncation function of [16, §6] sets $\mu(P) = e_P$ at type-2 points, so the divisor is $D = \eta^*(1)$; by (12), $\deg D = \deg \eta > 2g - 2$ and Riemann–Roch gives

$$(13) \quad N = h^0(D) = \deg \eta + 1 - g = g - 1 + r_0 + r_1 + r_\infty.$$

The only dependence on e_P is $r_1 = \deg \eta / e_P$, which cancels against $\deg D = e_P r_1 = \deg \eta$ identically — no Riemann–Roch is needed to see the cancellation. So N is unchanged at $e_P = 3$.

(C2) *The slope-0 count.* The Deuring–Shafarevich formula [6] pins the number of slope-0 segments at N , and [16, Corollary 7.14] cancels $r_0 + r_1 + r_\infty - |S|$ of them, leaving $g - 1 + |S|$: exactly the ramification-defined count of (1), *independent of e_P* . The target polygon is therefore the Kramer–Miller one and not a weakened variant.

- (C3) *The slope-1 half.* Bounds of the type of [16, Corollary 6.8] give the polygon below slope 1; [14, §7.3] completes it by a degree count — “from the Euler–Poincaré formula we know $L(\rho, s)$ has degree $2(g - 1 + m) + \sum (s_{\tau_i} - 1)$ ” — which is the cheap and correct thing to quote here. Poincaré duality enters [14] only for endpoint equality.
- (C4) *The functional analysis.* The Riemann–Roch step of [14, Proposition 7.2] is carried out on the *unweighted* space by reduction modulo $\mathfrak{m}$, and the weight does not appear in it at all. The weight enters only through the *diagonal* change of basis of their Proposition 7.4, under which every principal minor — hence the whole Fredholm series — is invariant, for any weight, provided both operators are nuclear. Nuclearity is $d(k) \rightarrow \infty$, which is Theorem 2.8.
- (C5) *The one strict inequality.* The single strict inequality in the perturbation machinery, [16, Definition 7.3(1)], is supplied by the margin $[kp - k(p - 1)]/\delta_P = k/\delta_P$, which is independent of p . The factor $p - 1$ that vanishes at $p = 2$ is not the source of that strictness. Likewise the gain $\pi_s^{n(p-1)}$ normalises to slope n/s at every p because $v_p(\pi) = 1/(p - 1)$, so the $p - 1$ cancels exactly. The only additive tightness anywhere in the local estimate is $d(6) = 1$, exactly at the threshold and exactly sufficient.
- (C6) *The wild points.* No step of the analysis at points of S carries a parity hypothesis; the splitting function is a product of rank-one Artin–Hasse factors with no cross terms [14, Proposition 5.5], and $\Omega_\rho = 0$ automatically for ρ of p -power order. The decay rate of that splitting function is at the theoretical ceiling at every prime [12, Theorems 12.6.4 and 19.4.1], [21, Proposition 2.12], so there is no $p = 2$ loss to recover there and none to lose.

Not re-verified here, and cited as their users cite them: the Deuring–Shafarevich formula [6], the local-to-global extension theorem of Katz and Gabber [11], Monsky’s trace formula [19, 20], Elkik [8], the functional analysis of [14, §6], and the internal proofs of [13] and [23].

5.3. Why Theorem 1.1 carries no truncation. Theorem 1.2 is capped and Theorem 1.1 is not, and the reason is (C4) together with the fourth row of the dictionary.

The cap on Theorem 1.2 originates in the exact sequence of [16, §6.2]. Making that sequence available for the repaired weight requires the growth parameter $m_{e,P} \geq 1$, hence $e \leq 1$ in the normalisation $m_{e,P} = 1/e$, which is what §3.1 converts into q -adic $r \leq 2^{1-n}$. The formalism that constraint lives in is the tuple m_e and the basis B^{m_e} .

[14] has no such formalism at the type-2 points. Its local growth module is $\mathcal{D} = \prod_n p^{b(n)} t^n \mathcal{O}_L$, coefficientwise, with integer exponents and no radius parameter; the Riemann–Roch step is weight-free; the weight enters only through a diagonal similarity. There is no free parameter for the exact sequence to constrain, and the requirement consumed in [14, §7.2, case (II)]

is the absolute statement $d(k) \geq 1$ in v_p . So the exactness statement does not arise on this route, and neither does the cap.

5.4. Two defects in the sources, repaired. Neither is a $p = 2$ obstruction; both are p -uniform, and both are vacuous for the characters of Theorem 1.1, for which $\varepsilon_Q = 0$. We record them because they are load-bearing for [14]’s general statement.

Proposition 5.1. *The bound $-q(e, j) \leq a(p - 1)$ asserted in [14, §4.1.1] is false, at every p ; witnesses are $(p, a, \varepsilon, j) = (2, 3, 3, 1)$, giving $-q = 4 > 3$, and $(3, 2, 5, 1)$, giving $-q = 5 > 4$. The weaker bound $-q \leq ap$ also fails — $(2, 4, 7, 3)$ gives $-q = 9 > 8$ — and neither can be repaired by shrinking s , both terms scaling as $1/s$. The correct bound is*

$$-q(e, j) \leq (p - 1) \frac{a(a + 1)}{2},$$

which equals $a(p - 1)$ only for $a = 1$.

Proposition 5.2. *The choice of s_Q required by [14, eq. (18)], namely an s_Q with $1/s_Q - \omega_Q/(as_Q(p - 1)) \geq 1$, is feasible, though the source does not argue it. Indeed ε_Q is the unique integer between 0 and $q - 2$ with the given digit sum, and the only $\varepsilon \leq p^a - 1$ with digit sum $a(p - 1)$ is $p^a - 1 = q - 1$, which is excluded; so $\omega_Q \leq a(p - 1) - 1$ and $1 - \omega_Q/(a(p - 1)) \geq 1/(a(p - 1)) > 0$, whence any sufficiently small s_Q works. At $p = 2$ this reads $\omega_Q \leq a - 1$.*

Both were checked exhaustively for $p \in \{2, 3, 5\}$ and $a \leq 6$, over all ε ; the sweeps are in the replication package.

Two smaller textual points, recorded so that a reader comparing displays does not stumble. [16, Definition 6.3] defines its module with a *strict* inequality $v_\pi(b_k) > k/m_P$ and then asserts that $\{\pi^{a(k)/m_P} t_P^{-k}\}$ is a formal basis of it, which with $>$ is false, the basis elements sitting exactly on the boundary; their own §2.1 uses $\geq$, and $\geq$ is the repair. Nothing downstream sees the difference, every estimate here being coefficientwise, and the point does not arise on the [14] side at all. Separately, the truncation function of [14, eq. (33)] reads $\mu(Q) = p$ at type-2 points where [16, eq. (11)] reads $p - 1$: the two denote the *same* truncation, one counting the first kept index and the other the last dropped one, and with $e_P = 3$ both read “drop poles of order ≤ 3 ”, which is (A1).

5.5. Proof of the main theorem.

Proof of Theorem 1.1. Let $q = 2^a$, $X/\mathbb{F}_q$ smooth affine, and ρ nontrivial of order 2^n . By Proposition 4.7 we may replace $\mathbb{F}_q$ by $\mathbb{F}_{q^2}$ if a is odd, so assume $3 \mid q - 1$; the character remains nontrivial because it is wildly ramified at some point of S and ramification is geometric, and neither polygon changes.

By Lemma 4.1 with $e = 3$ — available by Proposition 4.5 — there is a tame Belyi map $\eta : X \rightarrow \mathbb{P}_{\mathbb{F}_q}^1$ with $\eta(S) = \{0\}$ and uniform ramification index 3 over 1, satisfying (12). This is the input [14, Lemma 3.1] supplies at

$p \geq 3$; by Remark 4.2 the mixed fibre over 0 is admissible, and by Remark 4.3 the point over ∞ creates no further type-2 point.

At each P with $\eta(P) = 1$ the local situation is that of §2.1 with $e = 3$, and by Theorem 2.8 the weight (10) satisfies (A1)–(A5) with $d(k) \geq 1$ for every $k > 3$ and $d(k) \rightarrow \infty$. By §5.1 this is precisely the hypothesis that [14, §4.2 and Proposition 4.2] supply at $p \geq 3$, in that paper’s own coefficientwise module and with its own normalisation, the calibration being exact there.

The remaining hypotheses of [14] are p -uniform by (C1)–(C6): the truncation and the count (13) are unchanged at $e_P = 3$; the slope-0 multiplicity reduces to $g - 1 + |S|$ independently of e_P ; the slope-1 half comes from the degree count; the Fredholm series is invariant under the diagonal change of basis for any weight, and is nuclear because $d(k) \rightarrow \infty$; the one strict inequality has a p -independent margin; and the analysis at the wild points carries no parity hypothesis, with $\Omega_\rho = 0$ automatic for ρ of 2-power order. Running [14] with these two substitutions therefore yields $\text{NP}_q(\rho) \geq \text{HP}_q(\rho)$ with the polygon (1). By §5.3 the argument carries no truncation parameter, so the conclusion is the full polygon and holds for every n . $\square$

Proof of Theorem 1.2. The same two substitutions are made in [16] instead: its Proposition 4.3 by Lemma 4.1, and its Definition 6.3, Lemma 6.2, Proposition 6.4 and Remark 6.5 by Theorem 2.8. Its §§2–3, 4.2–4.4, 5, the wild part of 6.1, and 7.1–7.4 are p -uniform, and its Lemma 7.11, the unique consumer of $d(k) \geq 1$, is unchanged given Theorem 2.8: with $m_{e,P} = 1/e$ the columns at $\eta(P) = 1$ have slope $d(k)e$, so the requirement is $d(k) \geq 1$ non-strictly — correctly so, since $I^{<r}$ is defined with a strict inequality and a column of slope exactly r already lies outside it. The range is then as computed in §3.1: the exact sequence of [16, §6.2] requires $m_{e,P} \geq 1$, giving π -adic $r \leq 1$, hence q -adic $r \leq 2^{1-n}$ since $v_\pi(2) = 2^{n-1}$. $\square$

We flag once more what Theorem 1.2 rests on that Theorem 1.1 does not. The exact sequence of [16, §6.2] is asserted there without proof, at every p and for their own weight; only the $A^\dagger$ version is proved. It is a pre-existing gap in the source rather than a cost of the repair (Remark 2.7), and by §5.3 it does not arise on the route of Theorem 1.1 at all.

6. OPEN PROBLEMS

The full contact theory. Theorem 1.2 covers q -adic $r \leq 2^{1-n}$, and by Corollary 3.3 no choice of auxiliary tame index enlarges that range. The two routes that would — proving the exact sequence of [16, §6.2] outright, or re-running its §6.2 and §7 in the coefficientwise formulation of [14] — are set out in Remark 3.4, together with a third that would relax the cap to $r \leq 1/2$ uniformly in n if the rest of the argument survives a change of growth parameter. We emphasise that the first is a gap in the published source at every p , not something introduced here: only the $A^\dagger$ version of that

sequence is proved there, and by Remark 2.7 no admissible weight makes the relevant module a ring at any prime.

Deflation past the universal cap. Theorem 3.1 says the obstruction to a better rate is one explicit near-diagonal entry, of valuation exactly 1, contributed by the self-loop at $k = 2e$. That suggests stripping its span — and, if necessary, its finite forward orbit — as a finite-rank correction, factoring or bounding $\det(1 - Ms)$ as a small explicit block times a deflated determinant, and running the weight argument on the complement, where increments $d(k) \geq 2$ become available. The self-loop factor’s contribution is computable in closed form from Theorem 2.2, and it is where the polygon’s own slope- $\frac{1}{2}$ -type segment should sit. We state this as a proposal; we have neither a proof nor a computation behind it.

The weight theory at general odd e . Theorems 2.2 and 2.5, Lemma 2.6 and Theorem 3.1 hold for every odd e , with the sharpness threshold $1/(2e)$. Only the mod-6 case analysis of Theorem 2.8 is specific to $e = 3$; redoing it modulo $2e$ would make the extension-free variant of Remark 4.8, with $e = q - 1$, usable directly, though Proposition 4.7 already makes the $3 \mid q - 1$ condition free. A related and genuinely open question: whether a higher-degree auxiliary map could produce $e_P = 3$ over a field with $3 \nmid q - 1$.

Characters of general order. Theorem 1.1 is stated for ρ of 2-power order, which is where $\Omega_\rho = 0$ and the tame part is absent. Extending it to a general finite character in characteristic 2 means carrying the tame ramification data of [14, §4.1] through, and that is where Propositions 5.1 and 5.2 become load-bearing rather than vacuous. We have repaired both statements but not pursued the extension.

Replication. Every computational claim in this paper is backed by a program in the replication package, listed in the data-availability statement below. Each asserts its findings and exits nonzero on any failure. The package runs in two layers and the order is load-bearing. The primary layer is AXEYUM [4], pinned at

```
repository https://github.com/mjbommar/axeyum
branch      agent/noh-p2-axeyum-examples
commit      75663ef85c2dad4390a3b6d77361919a914642a9
```

where it runs the two self-checking examples `crates/axeyum-cas/examples/noh_u2_matrix.rs` and `noh_wt_certificate.rs`, after verifying by SHA-256 that the copies shipped with this paper are byte-identical to that commit. The second layer is an independent from-scratch reimplementation that shares no code with the first, and it covers ground the first does not — the linear-programming route to $1/6$, the orbit-sum weight, the Lubin–Tate invariance sweep, the Witt-level controls and the whole of Lemma 4.1 — just as the first covers the Dwork trace-formula anchor and the diagonal lattice certificate, which the second does not. Neither subsumes the other; a green run requires both. The package also carries the research

log: the source extractions, the adversarial audit that refuted or downgraded several intermediate claims, and an error ledger. A reader who wants to attack this paper will find the highest-yield targets named there — and the first of them is not any theorem above, but the non-load-bearing rows of the p -uniformity survey summarised in §5.2, which have been read once rather than re-derived.

DATA AND CODE AVAILABILITY

The self-checking verification programs for the local weight theory (the operator matrix and the Theorem 1–4 certificate), the from-scratch cross-checks, and the standalone verifier scripts accompany this paper in the repository <https://github.com/mjbommar/newton-over-hodge-char2> under `replication/`, together with the full research log. Each program asserts its findings and exits nonzero on any failure, so its exit status depends on what the run found rather than on the run completing. No external solver or computer-algebra system is required.

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The mathematics in this paper was produced by a multi-agent AI pipeline with an adversarial verification stage: proving agents wrote the arguments, an independent auditing agent re-derived them from the definitions and attempted to refute them, and every assertion carries the label the audit left it with. The complete audit trail, including the results that were refuted and withdrawn, is in `research-log/` in the accompanying repository. The author directed the program, is solely responsible for all content, and does not represent the machine-checked components as a substitute for human referee review.

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